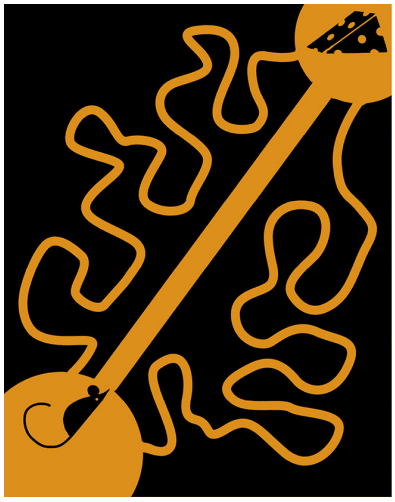


THE FITNESS OF A FORAGING ANIMAL DEPENDS ON ITS EFFICIENCY

OPTIMAL FORAGING THEORY

IN CONTEXT

KEY FIGURES

H. Ronald Pulliam (1945–), **Graham Pyke** (1948–), and **Eric Charnov** (1947–)

BEFORE

1966 John Merritt Emlen, Robert MacArthur, and Eric Pianka outline the concept of optimal foraging in two articles published in the *American Naturalist* magazine.

AFTER

1984 Argentinian–British zoologist Alejandro Kacelnik researches the foraging behavior of starlings to illustrate the marginal value theorem (MVT).

1986 Belgian ecologist Patrick Meire investigates prey selection by oystercatchers.

1989 Swiss environmental scientists T.J. Wolfe and Paul Schmid-Hempel examine how the weight of nectar carried by bees has an effect on the bees' foraging behavior.

Every plant and animal on Earth needs resources to survive. Plants obtain their nutrients and water from soil, and sunlight provides the energy for